

Blacklist Tracker Changelog

Full version history of Blacklist Tracker methodology decisions, from v1.0 to v3.993.

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LATEST VERSION

v3.993

May 12, 2026

FreezeWatch rebrand and primary-nav promotion

Promotes the Blacklist Tracker surface to a primary navigation hero under the FreezeWatch name, adding three issuer-sovereignty visualizations (Freezable Supply Meter, Intervention Seismograph, Sovereignty Lattice) above the existing intervention ledger. Event ingestion, status resolution, and the underlying coverage manifest are unchanged.

IMPACT SNAPSHOT

- Page moved from `/blacklist/` to `/freezewatch/` with a 301 redirect; sitemap priority raised to 0.85
- Primary navigation gains a 7th entry (`FreezeWatch`), replacing the `Blacklist` entry in the `TRACK` group
- User-facing copy on the surface refers to freeze interventions and issuer freezes; internal data model, table names, cron jobs, and API endpoints continue to use the `blacklist` term
- Three new client-side components render above the legacy intervention ledger: Freezable Supply Meter (sovereignty headline), Intervention Seismograph (quarterly cadence), Sovereignty Lattice (issuer × chain coverage grid)

v3.993

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Hide details

IMPACT NOTES

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- Analytics page identifier changed from `blacklist` to `freezewish` for filter, sort, and search events emitted by the page
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Commit provenance not recorded

v3.992

May 11, 2026

Public contract clarification

Clarifies the public blacklist contract without changing event ingestion: `/blacklist` status charts use the five-status model, CSV export covers the currently loaded table page, `/api/blacklist-summary` distinguishes legacy active-state fields from tracked freeze-ledger fields, coverage counts are manifest-derived, and chain identifiers are documented as `chainId` response/join fields rather than the `chain` display-name filter.

- The blacklist exposure model is documented as Yes, Dilutable, Upstream, Possible, and No
- Reserve, backing, custody, parent-asset, and CEX/custody-rail exposure resolve to Upstream through the any-reserve policy instead of Possible
- `/blacklist` CSV export is documented as the current server-returned page after filters, sorting, search, and pagination
- `/api/blacklist-summary` consumers are directed to `trackedAddressCount`, `trackedFrozenTotal`, and `trackedAmountGapCount`; legacy `active\*` fields remain for local net-active state-machine compatibility

Details

v3.991

May 5, 2026

Public transparency and snapshot semantics

Clarifies public API artifacts, UI provenance, and freeze-ledger snapshot semantics without changing event ingestion. `/api/blacklist` integration examples now use uppercase tracker symbols, frozen totals are labeled as last-known successful snapshots, and Tron missing-balance behavior is documented as provider-missing rather than false zero.

- OpenAPI/Postman document `/api/blacklist?stablecoin=` as an uppercase blacklist-tracker symbol filter
- Blacklist UI/CSV surfaces amount source/status plus contract/config/event provenance
- Public frozen totals are presented as last-known successful freeze-ledger snapshots, with provider failures preserving the previous successful value
- New freeze-ledger snapshots are described as contract/config scoped, with legacy symbol/chain/address fallback for older rows

Details

v3.99

Apr 21, 2026

Same-run Tron ledger reconciliation

Reapplies the Tron freeze-ledger mirror after current-balance cache writes so newly ingested Tron blacklist rows resolve in the same `sync-blacklist` cycle, and aligns admin recovery guidance away from destructive sync resets for generic amount-gap warnings.

- Fresh Tron blacklist/unblacklist rows no longer wait for the next 6-hour cron cycle before `current\_balance\_snapshot` amounts appear on matching event rows
- The admin current-balance backfill action now re-applies the Tron ledger mirror after rebuilding `blacklist\_current\_balances`
- Blacklist-gap recommendations now prioritize balance backfill, sync-state inspection, and targeted remediation ahead of `reset-blacklist-sync`

Details

v3.98

Apr 20, 2026

USDA destroy-event correction

Narrows Avalon USDe event tracking to the two events its verified contract actually emits: AddedBlackList(address) and RemovedBlackList(address). USDe remains directly freezable and role-burnable, but it no longer subscribes to Tether's DestroyedBlackFunds(address,uint256) topic because the verified USDe ABI/source does not expose that event.

- USDA blacklist/unblacklist event ingestion remains unchanged
- The report-card and blacklistability metadata still mark USDe as directly freezable because isBlackListed gates transfers and manager-controlled add/remove blacklist functions exist
- USDA no longer advertises or scans a non-existent DestroyedBlackFunds event in CONTRACT_CONFIGS
- Role-gated burn(address,uint256) is documented as a privileged destroy capability, but not mapped to blacklist-tracker destroy rows without a dedicated destroy event

Details

v3.97

Apr 19, 2026

Status amount-gap tolerance

Keeps isolated recent blacklist amount gaps below the incident threshold so a single unresolved provider/parser miss among thousands of events remains visible in diagnostics without degrading `/status` or `/admin` health.

- Recent amount gaps now need at least 5 rows in the 24-hour monitoring window before data quality degrades
- The stale threshold remains unchanged at 25 recent gaps or a 2% missing-amount ratio
- The ratio-based degraded threshold remains 1%, so broad amount-attribution failures still surface promptly

Details

v3.96

Apr 18, 2026

Gentle amount-gap recovery acceleration

Unblocks the EVM amount recovery lane by excluding Tron rows that are owned by the separate Tron ledger mirror, and raises the per-run recovery cap from 50 to 100 rows while staying inside the 6-hour sync cadence, D1 batch helper, rate limiters, and 900-subrequest run budget.

- EVM historical amount gaps no longer wait behind recent Tron pending rows in the per-row recovery selection
- Amount recovery now processes up to 100 rows per 6-hour sync-blacklist run instead of 50
- The acceleration remains bounded by the existing 7-minute sync runtime budget, 900-subrequest cap, and shared D1 batch chunking

Details

v3.95

Apr 17, 2026

Tier-1 coverage expansion

Adds eleven new tracked stablecoins across four new event families and six existing families. TUSD joins through a new TRUEUSD_EVENT_FAMILY that discriminates blacklist vs unblacklist via the bool at data slot 0 through the new BlacklistEventDef.eventTypeFromDataBoolIndex extension. JPYC adds CENTRE_BLOCKLISTED_FAMILY (Blocklisted/UnBlocklisted), FRXUSD adds FRAX_FREEZE_FAMILY (AccountFrozen/AccountThawed with non-indexed address), EURCV adds SOCGEN_FREEZE_FAMILY (batch AddressesFrozen/AddressesUnFrozen), NUSD adds NEUTRL_DENYLIST_FAMILY, and FIDD adds FIDELITY_RESTRICTION_FAMILY (TransferRestrictionImposed/Removed). USDA joins through legacy AddedBlackList/RemovedBlackList signatures, while USAT, AEUR, XUSD, and XAUM reuse existing families. Several BSC/Avalanche/Base deployments remain deferred pending block-explorer API access, and apxUSD is intentionally deferred because the verified ABI exposes no direction discriminator.

- Added TUSD on Ethereum (TRUEUSD_EVENT_FAMILY with bool-discriminator Blacklisted(address,bool) + DestroyedBlackFunds)
- Added EURCV on Ethereum (SOCGEN_FREEZE_FAMILY: batch AddressesFrozen(address[]) / AddressesUnFrozen(address[]))
- Added NUSD on Ethereum (NEUTRL_DENYLIST_FAMILY: AddedToDenylist / RemovedFromDenylist)
- Added USDA on Ethereum using legacy AddedBlackList / RemovedBlackList signatures; USAT on Ethereum reusing USDT0 family; AEUR on Ethereum reusing DUAL_INDEX_FREEZE family

Details

v3.94

Apr 17, 2026

Correctness + efficiency + minor coverage gaps

Worker correctness fixes (Gnosis dRPC 9k-block scan cap, dual-index-freeze family split from WLF1 destroy events, TronGrid failure propagation to the circuit breaker, EURC mirror-zero rows locked to permanently_unavailable, address[] expansion capped, Tron maxBlock cursor init), five D1 migrations (derived-row reset, sync_state dedup, EURC mirror-zero stamp, blacklist_events index adjustments, amount-status backfill), summary endpoint rewritten to aggregate in SQL with post-fetch counters inlined, frontend polish (data-driven stats strip, amount-status badge, per-coin stat border, CSV column

split, page-clamp + filter-reset coverage), and minor chain-coverage additions (Polygon USDQ, Arbitrum AID, Polygon TGBP; Base AID plus Base/BSC TGBP remain deferred).

- Gnosis BRZ scans now stay within the dRPC free-tier 10k-block cap and drain the ~12.5M-block backlog that previously masked 2 missed events
- TronGrid outages now throw through the per-config circuit breaker instead of silently returning empty arrays
- FDUSD / EURI / AEUR configs use DUAL_INDEX_FREEZE_EVENT_FAMILY without inheriting WLF1 FrozenAccountDrained / FrozenFundsReallocated topics they cannot emit
- EURC rows flagged as circle_mirror_zero_balance are stamped amount_status='permanently_unavailable' and no longer re-enter the recoverable-pending backfill pool

Details

v3.93

Apr 16, 2026

Backlog-safe scanner guardrails

Keeps EVM sync cursors pinned when the subrequest budget is exhausted before every configured event topic is scanned, marks budget-exhausted runs as degraded, and lowers the duplicate-row check chunk below D1's practical SQL-variable ceiling.

- Partial multi-topic EVM scans no longer advance cursors past unscanned topics
- Duplicate-row checks use smaller D1-safe chunks for high-row event batches
- Runs that skip configs after exhausting the subrequest budget now surface as degraded instead of ok

Details

v3.92

Apr 16, 2026

Amount attribution: Tron ledger mirror + derived-zero recovery

Introduces current_balance_snapshot amount source for Tron freeze-ledger mirroring, widens backfill to include legacy derived-zero EVM rows, and adds provenance badge for snapshot-sourced amounts.

- Tron USDT blacklist/unblacklist rows receive amounts from the freeze-ledger snapshot (~6,921 rows)
- Snapshot badge renders on amounts sourced from current_balance_snapshot
- Legacy derived-zero EVM rows re-enter the backfill pool (~4,381 rows)

Details

v3.91

Apr 16, 2026

Coverage quality + USDP addition

Extends wlfri-freeze family with FrozenAccountDrained/FrozenFundsReallocated destroy events (USD1/U/FDUSD/EURI), adds USDP Paxos coverage on Ethereum, and corrects Arbitrum FDUSD/AUSD/BUIDL startBlocks.

- Added FrozenAccountDrained + FrozenFundsReallocated destroy tracking for USD1/U/FDUSD/EURI
- Added USDP (Pax Dollar) Ethereum coverage via Paxos freeze family (65 existing events)
- Corrected Arbitrum startBlocks for FDUSD/AUSD/BUIDL to actual deployment blocks

[Details](#)

v3.9

Apr 15, 2026

Direct EVM coverage wave

Adds direct EVM blacklist/freeze coverage for FDUSD, BRZ, AUSD, MNEE, EURI, USDQ, USDO, USDX, AID, TGBP, and EURC, plus BUIDL seize-only tracking. Introduces indexed amount-topic parsing, explicit data-slot extraction for omnibus seize events, EURC mirror-zero suppression, and expanded non-USD valuation.

- Added FDUSD Ethereum/BSC/Arbitrum dual-index Freeze/Unfreeze tracking
- Added BRZ Ethereum/Gnosis, AUSD Arbitrum/Base, EURI Ethereum/BSC, USDQ Ethereum, USDO Ethereum/Base, USDX Ethereum, AID Ethereum, TGBP Ethereum/Avalanche, and EURC Ethereum/Base/Avalanche event coverage
- Added MNEE Ethereum freeze/unfreeze plus confiscation/burn tracking with indexed amount extraction
- Extended price-cache USD conversion to EURC, BRZ, EURI, and TGBP

[Details](#)

v3.8

Apr 15, 2026

First-wave CeFi coverage expansion

Expands blacklist/freeze tracking to first-wave centralized issuers with verified event surfaces: USDG, RLUSD, U, USDtb, and A7A5. Adds batch address-array parsing and non-USD A7A5 price-cache valuation while keeping clawback, allowlist/KYC, and non-EVM surfaces out of this release.

- Added USDG Ethereum FreezeAddress/UnfreezeAddress/FrozenAddressWipeAccountPaused/AccountUnpaused coverage using the Paxos pattern
- Added U Ethereum and BSC Freeze/Unfreeze coverage using the dual-indexed address pattern
- Added RLUSD Ethereum AccountsBlocked/AccountsUnblocked coverage; clawback remains out of scope until transaction-input classification exists
- Added USDtb Ethereum AccountsBlocked/AccountsUnblocked coverage with one row per address in batch events

[Details](#)

v3.7

Apr 8, 2026

Balance recovery accuracy and provider resilience

Remediates 16 audit findings across the balance recovery pipeline, freeze-ledger cache, and aggregation layer. Eliminates silent wrong-data paths, adds Ethereum mainnet dRPC/chain-RPC fallback, and fixes gold stablecoin USD conversion in all enrichment paths.

- Invalid block tags now return null instead of silently querying latest balance (Critical)
- Ethereum mainnet historical balance lookups now fall through dRPC and chain-RPC before Etherscan (Critical)
- Tron REST API returns null for missing token entries instead of false zero (Major)
- PAXG/XAUT events now receive USD conversion in enrichment and backfill paths (Major)

[Details](#)

v3.6

Mar 27, 2026

Freeze-ledger quarter attribution for the public chart

The public blacklist chart now buckets the persistent freeze ledger by blacklist quarter instead of summing raw event-time blacklist rows, so the quarterly bars explain the same tracked frozen total shown in the summary cards.

- The `/api/blacklist-summary`` chart now draws from ``blacklist_current_balances`` rather than raw ``blacklist_events`` intake amounts
- Each tracked balance is attributed to the latest recorded blacklist event for the same stablecoin/chain/address identity so re-blacklisted rows follow the active freeze cycle represented in the ledger
- Rows without a local blacklist timestamp fall back to the latest related event timestamp, then snapshot observation time, so tracked ledger value is not silently dropped from the chart

[Details](#)

v3.5

Mar 27, 2026

Persistent freeze-ledger snapshots and bootstrap reconciliation

The public frozen-total summary now uses a persistent freeze ledger instead of treating snapshot balances as a live current-state cache. Historical ETH/USDC, ETH/USDT, and TRON/USDT freeze rows were reconciled from the kyc.rip / stables.rip bootstrap so seized-and-burned balances remain visible after later unblacklist or destroy actions.

- Added tracked freeze-ledger metrics (``trackedAddressCount``, ``trackedFrozenTotal``, ``trackedAmountGapCount``) to blacklist summary responses
- Destroy events now persist their seized amount into the freeze ledger so burned balances remain counted
- Snapshot rows are now preserved across later unblacklist events instead of being deleted as if they were only live current balances
- Historical freeze-ledger bootstrap was reconciled against the external kyc.rip / stables.rip dataset for ETH USDC, ETH USDT, and TRON USDT

[Details](#)

v3.4

Mar 27, 2026

Active frozen-total ledger and Tron current-balance separation

Blacklist summary now distinguishes event-time amounts from active frozen balances, adds a dedicated current-balance cache for active blacklist records, and stops treating legacy Tron derived event amounts as authoritative history.

- Added ``blacklist_current_balances`` for current active blacklist balance snapshots
- Active Tron totals now prefer current TRC20 balances for active blacklist rows and destroy-event amounts when funds were seized and burned
- Blacklist summary gained active-record metrics (``activeAddressCount``, ``activeFrozenTotal``, ``activeAmountGapCount``)
- Legacy Tron ``derived`` blacklist/unblacklist event amounts are reset instead of being reused as event-time history

[Details](#)

v3.3

Mar 24, 2026

pyUSD and USD1 blacklist tracking coverage

Extended blacklist tracker to cover pyUSD (PayPal/Paxos) on Ethereum and Arbitrum, and USD1 (World Liberty Financial) on Ethereum, BSC, and Tron. Introduced configurable address topic index for two-indexed-address events.

- Added pyUSD FreezeAddress/UnfreezeAddress/FrozenAddressWipedAddressTopicIndex=2 for dual-indexed events event tracking (Paxos PaxosTokenV2 pattern)
- Added USD1 Freeze/Unfreeze event tracking with
- EVM parser now supports configurable topic index for affected address extraction
- Tron parser extended with tronResultKey for non-standard event parameter names

Details

v3.2

Mar 24, 2026

Provenance-aware rows and explicit amount semantics

Blacklist rows now persist emitting-contract provenance and explicit native/USD amount status fields so reprocessing and public consumers no longer rely on implicit inference.

- Rows now store config/contract provenance plus event signature metadata
- Amount semantics split into token-native and USD-at-event fields with explicit source/status flags
- Gap monitoring now tracks recoverable attribution failures rather than nullable amounts alone

Details

v3.1

Feb 25, 2026

Reconstructed

API-error-aware sync cursor protection

EVM scanning now distinguishes API failure from genuine no-event ranges so cursors do not advance on unreliable reads.

- EVM log fetching differentiates API failures (`null`) from valid empty responses (`[]`)
- On API failure, sync state is held and retried on the next cycle instead of advancing
- Metadata now reports `apiErrors` for operational observability

Details

v3.0

Feb 25, 2026

Reconstructed

Indexer-lag safety margins for cursor advancement

Head advancement added explicit safety buffers to prevent permanently skipping late-indexed explorer events.

- EVM no-event advancement now uses `head - safetyMargin` instead of raw head
- Reduces permanent event loss when explorer indexing lags chain tip
- Tron no-event advancement now uses `now - 15m` instead of wall-clock `now`

Details

v2.2

Feb 18, 2026

Reconstructed

Precision and integrity hardening

Amount math and log parsing were hardened to reduce silent corruption and improve sync telemetry.

- Token amounts switched to BigInt-safe decimal conversion to avoid large-value precision loss
- Sync now emits structured run metadata (`itemCount`, `contractsSkipped`, budget usage)
- Malformed EVM logs (invalid block/timestamp) are discarded instead of being inserted

Details

v2.1

Feb 18, 2026

Reconstructed

Pre-block balance sampling and zero-amount recovery

Balance attribution moved to pre-event block semantics, and backfill began explicitly reprocessing suspicious zero blacklist amounts.

- Balance enrichment samples `blockNumber - 1` for blacklist, unblacklist, and destroy
- Reduces same-block ordering artifacts that previously produced false zeros
- Backfill now re-attempts rows with `amount = 0` for blacklist events

Details

v2.0

Feb 12, 2026

Reconstructed

L2 balance reliability and budgeted full-scan loop

Major tracking architecture shift for L2 correctness and deterministic scan coverage under strict subrequest budgets.

- Introduced shared per-run subrequest budgeting with least-synced-first config ordering
- L2 balance sourcing evolved from Etherscan-only to RPC/dRPC archive-aware historical balance fetches
- Backfill moved ahead of incremental scan and EVM head caching reduced redundant rescans

Details

v1.2

Feb 11, 2026

Reconstructed

Coverage expansion: USDT0 and gold contract families

Expanded event coverage beyond legacy USDT/USDC patterns and fixed multiple cross-chain parsing mismatches.

- Added USDT0 event signatures and indexed-address parsing for upgraded Tether contracts
- Added PAXG and XAUT contract/event support with contract-specific mappings
- Per-contract decimals and Tron `0x`->`41` address normalization improved amount fidelity

Details

v1.1

Feb 11, 2026

Reconstructed

Ingestion-time enrichment and backfill foundation

Blacklist rows began storing balance context during ingestion, with a companion path for retroactive recovery of missing amounts.

- Blacklist/unblacklist rows are enriched with token balances before insert
- Backfill pipeline introduced for historical rows missing amount values
- Set groundwork for later destroy-event amount recovery hardening

Details

v1.0

Feb 9, 2026

Reconstructed

Initial Blacklist Tracker release

Launched multi-chain blacklist event ingestion, persistence schema, public API, and dashboard surface.

- Initial incremental EVM + Tron event sync for major fiat-backed stablecoins
- Introduced `blacklist\_events` and `blacklist\_sync\_state` tables
- Exposed tracker data through `/api/blacklist` and frontend event views

Details

WHITE PAPER

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